ARON BERGER

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Available for 4th Co-op Fall 2026 and 5th Co-op Summer 2027

EDUCATION

Bachelor of Science, Electrical Engineering August 2023 – April 2028
University of Cincinnati, Cincinnati, Ohio GPA: 3.95 / 4.00

- Relevant Coursework: Digital Design, Electronics I, Electronics II, Network Analysis, Signals and Systems, Semiconductor Devices, Engineer Design Thinking, Semiconductor Physics

WORK EXPERIENCE

Electronics Quality Engineer August 2024 – April 2026
Valco Melton, Cincinnati, Ohio

- Designed and implemented a PCB retrofit solution that enhanced an existing test harness, extending its functionality to support the latest cable variants.
- Overhauled a burn-in rack's 80-cable infrastructure, improving defect detection sensitivity and preventing subtle errors from reaching the field.
- Strengthened electrostatic discharge safety protocols by utilizing quantitative data and industry standards to devise solutions that helped resolve critical ISO audit findings.
- Automated data collection and formatting for monthly continuous-improvement meetings, reducing information analysis time by 70%.
- Engineered a scalable Microsoft Power Query solution to automate parts list aggregation and comparison across control unit families; replacing a manual process that degraded with complexity, improving assembly accuracy and consistency.

SKILLS & INTERESTS

Software: Python, C++, LabVIEW, SolidWorks, CAMI CableEye, Autodesk Inventor, AutoCAD, Multisim, LTspice
Hardware: Circuit design, Breadboarding, Arduino, Soldering, 3D printing
Industry Involvement: IEEE, Geauga Engineering and Robotics 4-H Club Judge

TECHNICAL PROJECTS

Analog Circuit Design and Analysis August 2025 – December 2025
University of Cincinnati, Cincinnati, Ohio

- Designed and tested common-emitter and common-source amplifier configurations across varying gain levels, demonstrating the critical role of gain control resistors in amplifier performance.
- Analyzed the effect of coupling and bypass capacitor selection on input and output waveforms, developing practical intuition for frequency response and signal integrity tradeoffs.
- Constructed a full-wave rectifier from discrete stages, gaining hands-on insight into how each stage contributes to a stable DC output.

Conway's Game of Life April 2024 – May 2024
University of Cincinnati, Cincinnati, Ohio

- Engineered an embedded implementation of Conway's Game of Life in C++, programming an Arduino Uno to drive an LED matrix display in real time.
- Modeled and analyzed system behavior through iterative simulation loops, leveraging cellular automaton logic to observe emergent patterns across dynamic states.